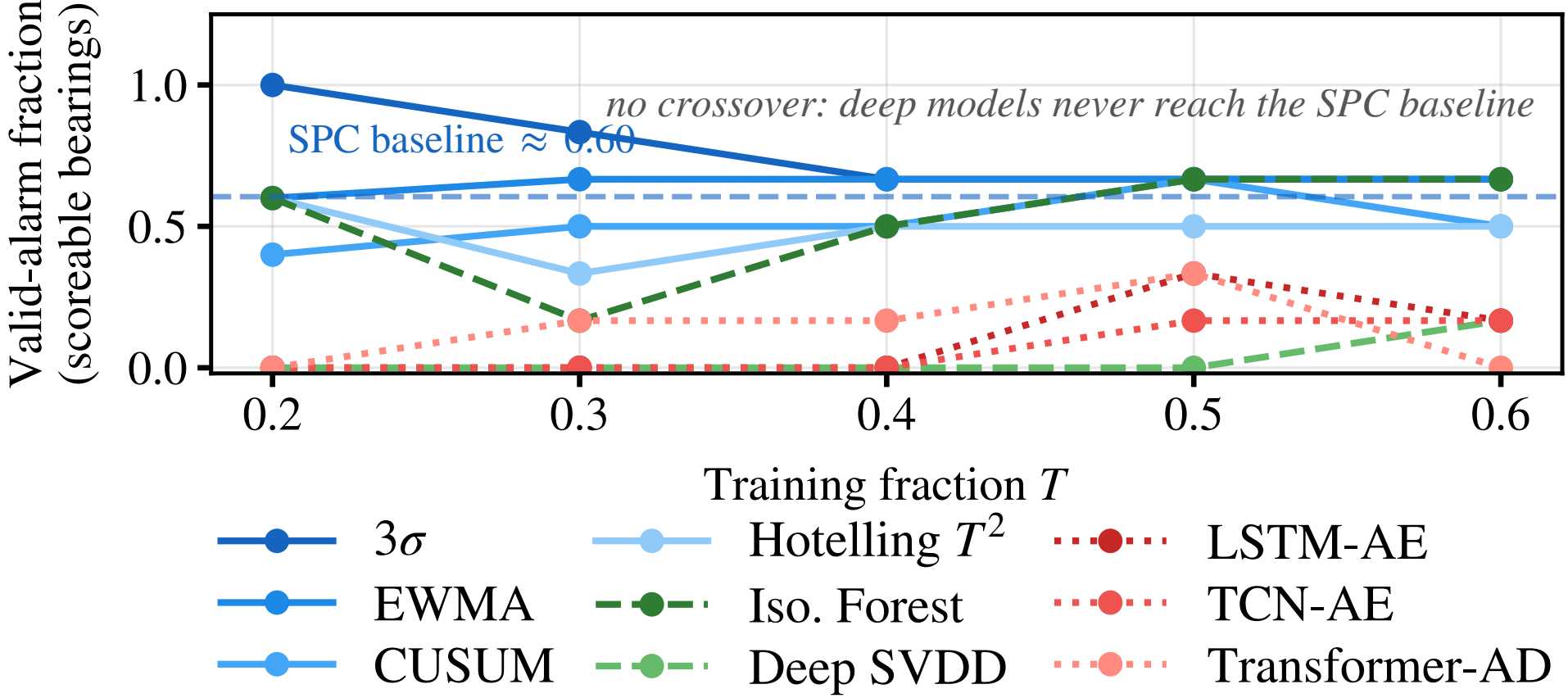
Valid-alarm fraction
(scoreable bearings)
1.0
0.5
0.0
SPC baseline ≈ 0.60
no crossover: deep models never reach the SPC baseline
0.2
0.3
0.4
0.5
0.6
Training fraction T
3σ
EWMA
CUSUM
Hotelling T^2
Iso. Forest
Deep SVDD
LSTM-AE
TCN-AE
Transformer-AD